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CHE 330 Hypothesis Testing II: F-Test / Analysis of Variance (ANOVA)[†] Fall 2011

The purpose of an analysis of variance or ANOVA is to find out whether data from several population groups of interest have a common mean, i.e., to determine whether the groups share a common measured characteristic or are, in fact, different in the measured characteristic. Thus, ANOVA is used to determine if there is any difference between the means of two or more population groups. If there are only two population groups, one can employ the simple hypothesis testing approach to determine if there is a statistical difference between the two means. When there are more than two population groups with more than two means to compare, one could attempt multiple pair-wise comparisons. Thus, for three population groups, A, B, and C, such an approach would yield three binary comparisons, namely, A-B, A-C, B-C. Likewise, for four population groups, A, B, C, and D, six binary comparisons would result, namely, A-B, A-C, A-D, B-C, B-D, and C-D. The number of binary comparisons increases rapidly with the number of population groups. The approach is inefficient and, more importantly, it suffers from two serious drawbacks that make the results less than reliable. The first drawback is an increased likelihood of the so-called *Type I error*. Recall that in a standard hypothesis test, the significance level is the probability that one might accidentally reject a true null hypothesis. When the number of binary comparisons increases, the chances of these *Type I errors* increases proportionally. The second drawback is a failure to account for possible *interaction effects* that cannot be detected or analyzed in a simple hypothesis test (see later).

An Illustrative Example on Analysis of Variance (ANOVA)

Samples were taken of milk shipment delivered by five different tanker trucks, and bacteria count analyses were performed with the following results:

Tanker 1	Tanker 2	Tanker 3	Tanker 4	Tanker 5
24	14	11	7	19
15	7	9	7	24
21	12	7	4	19
27	17	13	7	15
33	14	12	12	10
23	16	18	18	20

It is suspected that the bacteria count number varies with the delivery tankers. Use ANOVA to test the hypothesis at a 0.05 significance level and compare your results with that obtained using Excel's Single Factor Anova, which can be found under Tools and then Data Analysis.

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Solution

At first glance, the data seems to suggest that the bacteria count in milk is a function of delivery tanker. We need to confirm to confirm the suspicion with an ANOVA analysis.

First, postulate the Null Hypothesis that the above data merely reflects normal scatters in data and that there is no difference in the sample means of bacteria counts with tankers.

Mathematically, the Null Hypothesis may be formulated as

$$\mathbf{H}_0: \langle \mathbf{Y}_{1,j} \rangle = \langle \mathbf{Y}_{2,j} \rangle = \dots = \langle \mathbf{Y}_{5,j} \rangle$$

The Alternate Hypothesis may then be formulated as $\mathbf{H}_a: \langle \mathbf{Y}_{1,j} \rangle \neq \langle \mathbf{Y}_{2,j} \rangle \neq \dots \neq \langle \mathbf{Y}_{5,j} \rangle$

ANOVA provides a prescribed protocol to validate or reject the Null Hypothesis. The procedure is as follows (see later for a simpler procedure):

- (1) Calculate the *Level* mean of each group of data for each level of the factor (in this case, tanker) $\langle \mathbf{Y}_{i,j} \rangle = [\mathbf{Y}_{i,1} + \mathbf{Y}_{i,2} + \dots + \mathbf{Y}_{i,n_i}] / N_i$ where $N_i \equiv$ number of data points corresponding to the i^{th} level. Results are shown in the table on next page.
- (2) Calculate the *Grand* or *Overall* mean of ALL data $\langle \mathbf{Y} \rangle = [\sum (\sum \mathbf{Y}_{i,j})] / N_{\text{Total}} = [\sum (\sum \mathbf{Y}_{i,j})] / [\sum N_i]$
Note: Grand Mean = Average of Level Means.
- (3) Calculate the *Total Sum of Squares of Differences* of ALL data points.
 $\text{SSD}_T \equiv \{ \sum [\sum (\mathbf{Y}_{i,j} - \langle \mathbf{Y} \rangle)^2] \}_I = [\sum (\sum \mathbf{Y}_{i,j}^2)]_I - N_{\text{Total}} \langle \mathbf{Y} \rangle^2$. Results of Steps 2 & 3 are shown in the table on the page after next.
- (4) Calculate the *Sum of Squares of Differences Due to Different Levels of a Factor* (which measures the variability due to factors) $\text{SSD}_L \equiv \sum [N_i \langle \mathbf{Y}_{i,j} \rangle^2]_I - N_{\text{Total}} \langle \mathbf{Y} \rangle^2$
- (5) Calculate the *Sum of Squares of Differences Due to Random Errors* (which measures the variability due to unexplained factors)
 $\text{SSD}_E \equiv \text{SSD}_T - \text{SSD}_L = \sum [(N_i - 1) \text{St.Dev.}_i^2]_I$
- (6) Calculate the *Degrees of Freedom for the Total Data Points* (df_T), *for the Levels of a factor* (df_L), and *for Random Errors* (df_E) as follows: $df_T = N_{\text{Total}} - 1$, $df_L = N_L - 1$, $df_E = N_{\text{Total}} - N_L$ where N_L is the total number of levels for a factor.
- (7) Calculate the *Mean Square Value for Different Levels* of a factor and *for Errors* as defined below: $\text{MS}_L \equiv \text{SSD}_L / df_L = \text{SSD}_L / (N_L - 1)$, $\text{MS}_E \equiv \text{SSD}_E / df_E = \text{SSD}_E / (N_{\text{Total}} - N_L)$
- (8) Calculate the F-value for the Null Hypothesis as follows, $F = \text{MS}_L / \text{MS}_E$
- (9) Look up in an F-table the critical F-value for the specified significance level and degrees of freedom for the factors and errors, $F_{\text{critical}} = F_{\alpha, df_L, df_E}$
- (10) Finally, compare F with F_{critical} to make one of the following conclusions:
If $F > F_{\text{critical}}$, $\mathbf{H}_0$ may be rejected and $\mathbf{H}_a$ accepted at the specified significance level.
If $F < F_{\text{critical}}$, $\mathbf{H}_0$ may NOT be rejected and one must conclude that there is insufficient evidence to accept $\mathbf{H}_a$ at the specified significance level.

The hand-calculated ANOVA results are given below:

	Tanker 1	Tanker 2	Tanker 3	Tanker 4	Tanker 5
Bacteria	24	14	11	7	19
Count= $\mathbf{Y}_{i,j}$	15	7	9	7	24
	21	12	7	4	19
	27	17	13	7	15

	33	14	12	12	10	
	23	16	18	18	20	
Mean_I <Y_{I,j}>	23.8	13.3	11.7	9.2	17.8	
St. Dev._I	6.0	3.6	3.8	5.0	4.8	Sum
N_I*Mean_I²	3408.17	1066.67	816.67	504.17	1908.17	7703.83
(N_I-1)StDev_I²	180.8	63.3	71.3	126.8	114.8	557.17

SSD_T = Grand SSD = 1360.17 (see next table) or, alternatively,

$$SSD_T = [\sum(\sum Y_{I,j}^2)]_I - N_{Total} <Y>^2 = \text{Sum}(Y_{I,j}^2) - N_{Total} <Y>^2$$

$$SSD_T = 8261 - 30*15.17^2 = 1360.17 \text{ (see next table)}$$

$$SSD_L = \sum[N_I <Y_{I,j}>^2]_I - N_{Total} <Y>^2 = 7703.83 - 30*15.17^2 = 803 \text{ (see above \& next table)}$$

$$SSD_E = SSD_T - SSD_L = 557.17 \text{ or, alternatively, } SSD_E = \sum[(N_I - 1)St.Dev._I^2]_I = 557.17$$

$$MS_L = SS_L/(5-1) = 200.75 \quad MS_E = SS_E/(30-5) = 22.29 \quad F = MS_L/MS_E = 9.01$$

From F-tables, $F_{critical} = F_{0.05, 4, 25} = 2.758$ at a 0.05 significance level

All Data

Y_{I,j}	Y_{I,j}²
24	576
15	225
21	441
27	729
33	1089
23	529
14	196
7	49
12	144
17	289
14	196
16	256
11	121
9	81
7	49
13	169
12	144
18	324

	7	49	
	7	49	
	4	16	
	7	49	
	12	144	
	18	324	
	19	361	
	24	576	
	19	361	
	15	225	
	10	100	
	20	400	
Grand Mean			
<Y>	15.17	8261	Sum($Y_{i,j}^2$)
Grand St			
Dev s	6.85		
Grand			
Variance s^2	46.90		
Grand SSD	1360.17		

Since $F = 9.01 > F_{\text{critical}} = 2.758$, we can reject H_0 (the Null Hypothesis that the sample mean is uniform for all tankers) and accept H_a (the Alternate Hypothesis that the sample mean varies with the tanker) *at the 0.05 significance level*. The same conclusion can also be made at a 0.01 significance level because $F = 9.01 > F_{\text{critical}} = F_{0.01, 4, 25} = 4.177$.

The results of Excel's single factor ANOVA are presented below:

Anova: Single Factor

SUMMARY			
Groups	Count	Sum	Average
Column 1	6	143	23.833333
Column 2	6	80	13.333333
Column 3	6	70	11.666667
Column 4	6	55	9.166667
Column 5	6	107	17.833333

ANOVA			
Source of Variation	SS	df	MS
Between Groups	803	4	200.75
Within Groups	557.1667	25	22.286667
Total	1360.167	29	

It is seen that the hand-calculated and Excel results are in agreement, as to be expected.

Alternate, Simpler Hand-Calculation Solution

A simpler ANOVA procedure to validate or reject the Null Hypothesis is as follows:

- (1) Set up the table of data with data entries on the various levels for the factor.
- (2) Compute the mean of data within each level (Level Mean, LM) using the **average** (___ : ___) feature on *Excel*.
- (3) Compute the variance within each level (**Level Variance**, s_L^2) using the relation, **Level Variance** $s_L^2 = (\text{Level Standard Deviation } s_L)^2 = (\text{stdev}(\text{___ : ___}))^2$ feature on *Excel*.
- (4) Compute MS_E by setting it equal to the **average of the level variances**, i.e.,

$$MS_E = \frac{1}{n_L} \sum_{j=1}^{n_L} (s_L^2)_j$$
using the **average** (___ : ___) feature on *Excel*.
- (5) Compute the **variance of the level means**, $s_{LM}^2 = (\text{Level Mean Standard Deviation } s_{LM})^2 = (\text{stdev}(\text{___ : ___}))^2$ feature on *Excel*.
- (6) Compute MS_L by setting it equal $n_I s_{LM}^2$, i.e., $MS_L = n_I s_{LM}^2$, where n_I is the number of data entries within each level.
- (7) Then compute $F = MS_L / MS_E$ and compare it against F_c corresponding to the specified significance level.

The computation steps are summarized in the table below.

	Tanker i				
	Tanker 1	Tanker 2	Tanker 3	Tanker 4	Tanker 5
	24	14	11	7	19
	15	7	9	7	24
Bacteria Count = $Y_{I,j}$	21	12	7	4	19
	27	17	13	7	15
	33	14	12	12	10
	23	16	18	18	20
Level Mean, LM	23.833	13.333	11.667	9.167	17.833
Variance within Level, s_L^2	36.167	12.667	14.267	25.367	22.967
$MS_L = MS_B = n_I s_{LM}^2 =$	200.75				$F = MS_L / MS_E =$ 9.01

This alternate simpler method yields the same results as in the previous two cases.

Table 1. F-table for 0.05 significance level. df1 (top row) = df_F , df2 (left column) = df_E .

df2\df1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	22	24	26	28	30	35	40	45	50	60	70	80	100	200	500	1000	>1000	df1/df2		
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.76	8.74	8.73	8.71	8.70	8.69	8.68	8.67	8.67	8.66	8.65	8.64	8.63	8.62	8.62	8.60	8.59	8.59	8.58	8.57	8.57	8.56	8.55	8.54	8.53	8.53	8.54	3		
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.94	5.91	5.89	5.87	5.86	5.84	5.83	5.82	5.81	5.80	5.79	5.77	5.76	5.75	5.75	5.73	5.72	5.71	5.70	5.69	5.68	5.67	5.66	5.65	5.64	5.63	5.63	5.63	4	
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.70	4.68	4.66	4.64	4.62	4.60	4.59	4.58	4.57	4.56	4.54	4.53	4.52	4.50	4.50	4.48	4.46	4.45	4.44	4.43	4.42	4.42	4.41	4.39	4.37	4.37	4.36	4.36	5	
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.03	4.00	3.98	3.96	3.94	3.92	3.91	3.90	3.88	3.87	3.86	3.84	3.83	3.82	3.81	3.79	3.77	3.76	3.75	3.74	3.73	3.72	3.71	3.69	3.68	3.67	3.67	3.67	6	
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.60	3.57	3.55	3.53	3.51	3.49	3.48	3.47	3.46	3.44	3.43	3.41	3.40	3.39	3.38	3.36	3.34	3.33	3.32	3.30	3.29	3.29	3.27	3.25	3.24	3.23	3.23	3.23	7	
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.31	3.28	3.26	3.24	3.22	3.20	3.19	3.17	3.16	3.15	3.13	3.12	3.10	3.09	3.08	3.06	3.04	3.03	3.02	3.01	2.99	2.99	2.97	2.95	2.94	2.93	2.93	2.93	8	
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.10	3.07	3.05	3.03	3.01	2.99	2.97	2.96	2.95	2.94	2.92	2.90	2.89	2.87	2.86	2.84	2.83	2.81	2.80	2.79	2.78	2.77	2.76	2.73	2.72	2.71	2.71	2.71	9	
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.94	2.91	2.89	2.86	2.85	2.83	2.81	2.80	2.79	2.77	2.75	2.74	2.72	2.71	2.70	2.68	2.66	2.65	2.64	2.62	2.61	2.60	2.59	2.56	2.55	2.54	2.54	2.54	10	
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.82	2.79	2.76	2.74	2.72	2.70	2.69	2.67	2.66	2.65	2.63	2.61	2.59	2.58	2.57	2.55	2.53	2.52	2.51	2.49	2.48	2.47	2.46	2.43	2.42	2.41	2.41	2.41	11	
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.72	2.69	2.66	2.64	2.62	2.60	2.58	2.57	2.56	2.54	2.52	2.51	2.49	2.48	2.47	2.44	2.43	2.41	2.40	2.38	2.37	2.36	2.35	2.32	2.31	2.30	2.30	2.30	12	
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.63	2.60	2.58	2.55	2.53	2.51	2.50	2.48	2.47	2.46	2.44	2.42	2.41	2.39	2.38	2.36	2.34	2.33	2.31	2.30	2.28	2.27	2.26	2.23	2.22	2.21	2.21	2.21	13	
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.57	2.53	2.51	2.48	2.46	2.44	2.43	2.41	2.40	2.39	2.37	2.35	2.33	2.32	2.31	2.28	2.27	2.25	2.24	2.22	2.21	2.20	2.19	2.16	2.14	2.14	2.13	2.13	14	
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.51	2.48	2.45	2.42	2.40	2.38	2.37	2.35	2.34	2.33	2.31	2.29	2.27	2.26	2.25	2.22	2.20	2.19	2.18	2.16	2.15	2.14	2.12	2.10	2.08	2.07	2.07	2.07	15	
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.46	2.42	2.40	2.37	2.35	2.33	2.32	2.30	2.29	2.28	2.25	2.24	2.22	2.21	2.19	2.17	2.15	2.14	2.12	2.11	2.09	2.08	2.06	2.05	2.03	2.02	1.99	1.97	1.96	16
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.41	2.38	2.35	2.33	2.31	2.29	2.27	2.26	2.24	2.23	2.21	2.19	2.17	2.16	2.15	2.12	2.10	2.09	2.08	2.06	2.05	2.03	2.02	1.99	1.98	1.95	1.93	1.92	1.92	17
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.37	2.34	2.31	2.29	2.27	2.25	2.23	2.22	2.20	2.19	2.17	2.15	2.13	2.12	2.11	2.08	2.06	2.05	2.04	2.02	2.00	1.99	1.98	1.95	1.93	1.92	1.92	1.92	18	
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.34	2.31	2.28	2.26	2.23	2.21	2.20	2.18	2.17	2.16	2.13	2.11	2.10	2.08	2.07	2.05	2.03	2.01	2.00	1.98	1.97	1.96	1.94	1.91	1.89	1.88	1.88	1.88	19	
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.31	2.28	2.25	2.23	2.20	2.18	2.17	2.15	2.14	2.12	2.10	2.08	2.07	2.05	2.04	2.01	1.99	1.98	1.97	1.95	1.93	1.92	1.91	1.88	1.86	1.85	1.84	1.84	20	
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.26	2.23	2.20	2.17	2.15	2.13	2.11	2.10	2.08	2.07	2.05	2.03	2.01	2.00	1.98	1.96	1.94	1.92	1.91	1.89	1.88	1.86	1.85	1.82	1.80	1.79	1.78	1.78	22	
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.22	2.18	2.15	2.13	2.11	2.09	2.07	2.05	2.04	2.03	2.00	1.98	1.97	1.95	1.94	1.91	1.89	1.88	1.86	1.84	1.83	1.82	1.80	1.77	1.75	1.74	1.73	1.73	24	
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.18	2.15	2.12	2.09	2.07	2.05	2.03	2.02	2.00	1.99	1.97	1.95	1.93	1.91	1.90	1.87	1.85	1.84	1.82	1.80	1.79	1.78	1.76	1.73	1.71	1.70	1.69	1.69	26	
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.15	2.12	2.09	2.06	2.04	2.02	2.00	1.99	1.97	1.96	1.93	1.91	1.90	1.88	1.87	1.84	1.82	1.80	1.79	1.77	1.75	1.74	1.73	1.69	1.67	1.66	1.66	28		
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.13	2.09	2.06	2.04	2.01	1.99	1.98	1.96	1.95	1.93	1.91	1.89	1.87	1.85	1.84	1.81	1.79	1.77	1.76	1.74	1.72	1.71	1.70	1.66	1.64	1.63	1.62	30		
35	4.12	3.27	2.87	2.64	2.49	2.37	2.29	2.22	2.16	2.11	2.08	2.04	2.01	1.99	1.96	1.94	1.92	1.91	1.89	1.88	1.85	1.83	1.82	1.80	1.79	1.76	1.74	1.72	1.70	1.68	1.66	1.65	1.63	1.60	1.57	1.57	1.56	35		
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.04	2.00	1.97	1.95	1.92	1.90	1.89	1.87	1.85	1.84	1.81	1.79	1.77	1.76	1.74	1.72	1.69	1.67	1.66	1.64	1.62	1.61	1.59	1.55	1.53	1.52	1.51	40		
45	4.06	3.20	2.81	2.58	2.42	2.31	2.22	2.15	2.10	2.05	2.01	1.97	1.94	1.92	1.89	1.87	1.86	1.84	1.82	1.81	1.78	1.76	1.74	1.73	1.71	1.68	1.66	1.64	1.63	1.60	1.59	1.57	1.55	1.51	1.49	1.48	1.47	45		
50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13	2.07	2.03	1.99	1.95	1.92	1.89	1.87	1.85	1.83	1.81	1.80	1.78	1.76	1.74	1.72	1.70	1.69	1.66	1.63	1.61	1.60	1.58	1.56	1.54	1.52	1.48	1.46	1.45	1.44	50		
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.95	1.92	1.89	1.86	1.84	1.82	1.80	1.78	1.76	1.75	1.72	1.70	1.68	1.66	1.65	1.62	1.59	1.57	1.55	1.53	1.50	1.49	1.47	1.45	1.44	1.41	1.40	1.39	60	
70	3.98	3.13	2.74	2.50	2.35	2.23	2.14	2.07	2.02	1.97	1.93	1.89	1.86	1.84	1.81	1.79	1.77	1.75	1.74	1.72	1.70	1.67	1.65	1.64	1.62	1.59	1.57	1.55	1.53	1.50	1.49	1.47	1.45	1.40	1.37	1.36	1.35	70		
80	3.96	3.11	2.72	2.49	2.33	2.21	2.13	2.06	2.00	1.95	1.91	1.88	1.84	1.82	1.79	1.77	1.75	1.73	1.72	1.70	1.68	1.65	1.63	1.62	1.60	1.57	1.54	1.52	1.51	1.48	1.46	1.45	1.43	1.38	1.35	1.34	1.33	80		
100	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.97	1.93	1.89	1.85	1.82	1.79	1.77	1.75	1.73	1.71	1.69	1.68	1.65	1.63	1.61	1.59	1.57	1.54	1.52	1.49	1.48	1.45	1.43	1.41	1.39	1.34	1.31	1.30	1.28	100		
200	3.89	3.04	2.65	2.42	2.26	2.14	2.06	1.98	1.93	1.88	1.84	1.80	1.77	1.74	1.72	1.69	1.67	1.66	1.64	1.62	1.60	1.57	1.55	1.53	1.52	1.48	1.46	1.43	1.41	1.39	1.36	1.35	1.32	1.26	1.22	1.21	1.19	200		
500	3.86	3.01	2.62	2.39	2.23	2.12	2.03	1.96	1.90	1.85	1.81	1.77	1.74	1.71	1.69	1.66	1.64	1.62	1.61	1.59	1.56	1.54	1.52	1.50	1.48	1.45	1.42	1.40	1.38	1.35	1.32	1.30	1.28	1.21	1.16	1.14	1.12	500		
1000	3.85	3.00	2.61	2.38	2.22	2.11	2.02	1.95	1.89	1.84	1.80	1.76	1.73	1.70	1.68	1.65	1.63	1.61</																						

Table 2. F-table for 0.01 significance level. df1 (top row) = df_F , df2 (left column) = df_E .

df2\df1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	22	24	26	28	30	35	40	45	50	60	70	80	100	200	500	1000	>1000	df1/df2
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23	27.13	27.05	26.98	26.92	26.87	26.83	26.79	26.75	26.72	26.69	26.64	26.60	26.56	26.53	26.50	26.45	26.41	26.38	26.35	26.32	26.29	26.27	26.24	26.18	26.15	26.13	26.15	3
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.45	14.37	14.31	14.25	14.20	14.15	14.11	14.08	14.05	14.02	13.97	13.93	13.89	13.86	13.84	13.79	13.75	13.71	13.69	13.65	13.63	13.61	13.58	13.52	13.49	13.47	13.47	4
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.96	9.89	9.82	9.77	9.72	9.68	9.64	9.61	9.58	9.55	9.51	9.47	9.43	9.40	9.38	9.33	9.29	9.26	9.24	9.20	9.18	9.16	9.13	9.08	9.04	9.03	9.02	5
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.79	7.72	7.66	7.61	7.56	7.52	7.48	7.45	7.42	7.40	7.35	7.31	7.28	7.25	7.23	7.18	7.14	7.11	7.09	7.06	7.03	7.01	6.99	6.93	6.90	6.89	6.89	6
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.54	6.47	6.41	6.36	6.31	6.28	6.24	6.21	6.18	6.16	6.11	6.07	6.04	6.02	5.99	5.94	5.91	5.88	5.86	5.82	5.80	5.78	5.75	5.70	5.67	5.66	5.65	7
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.73	5.67	5.61	5.56	5.52	5.48	5.44	5.41	5.38	5.36	5.32	5.28	5.25	5.22	5.20	5.15	5.12	5.09	5.07	5.03	5.01	4.99	4.96	4.91	4.88	4.87	4.86	8
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.18	5.11	5.05	5.01	4.96	4.92	4.89	4.86	4.83	4.81	4.77	4.73	4.70	4.67	4.65	4.60	4.57	4.54	4.52	4.48	4.46	4.44	4.42	4.36	4.33	4.32	4.32	9
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.77	4.71	4.65	4.60	4.56	4.52	4.49	4.46	4.43	4.41	4.36	4.33	4.30	4.27	4.25	4.20	4.17	4.14	4.12	4.08	4.06	4.04	4.01	3.96	3.93	3.92	3.91	10
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.46	4.40	4.34	4.29	4.25	4.21	4.18	4.15	4.12	4.10	4.06	4.02	3.99	3.96	3.94	3.89	3.86	3.83	3.81	3.78	3.75	3.73	3.71	3.66	3.62	3.61	3.60	11
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.22	4.16	4.10	4.05	4.01	3.97	3.94	3.91	3.88	3.86	3.82	3.78	3.75	3.72	3.70	3.65	3.62	3.59	3.57	3.54	3.51	3.49	3.47	3.41	3.38	3.37	3.36	12
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	4.02	3.96	3.91	3.86	3.82	3.78	3.75	3.72	3.69	3.66	3.62	3.59	3.56	3.53	3.51	3.46	3.43	3.40	3.38	3.34	3.32	3.30	3.27	3.22	3.19	3.18	3.17	13
14	8.86	6.51	5.56	5.04	4.70	4.46	4.28	4.14	4.03	3.94	3.86	3.80	3.75	3.70	3.66	3.62	3.59	3.56	3.53	3.51	3.46	3.43	3.40	3.37	3.35	3.30	3.27	3.24	3.22	3.18	3.16	3.14	3.11	3.06	3.03	3.01	3.01	14
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.73	3.67	3.61	3.56	3.52	3.49	3.45	3.42	3.40	3.37	3.33	3.29	3.26	3.24	3.21	3.17	3.13	3.10	3.08	3.05	3.02	3.00	2.98	2.92	2.89	2.88	2.87	15
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.62	3.55	3.50	3.45	3.41	3.37	3.34	3.31	3.28	3.26	3.22	3.18	3.15	3.12	3.10	3.05	3.02	2.99	2.97	2.93	2.91	2.89	2.86	2.81	2.78	2.76	2.75	16
17	8.40	6.11	5.19	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.52	3.46	3.40	3.35	3.31	3.27	3.24	3.21	3.19	3.16	3.12	3.08	3.05	3.03	3.00	2.96	2.92	2.89	2.87	2.83	2.81	2.79	2.76	2.71	2.68	2.66	2.65	17
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.43	3.37	3.32	3.27	3.23	3.19	3.16	3.13	3.10	3.08	3.03	3.00	2.97	2.94	2.92	2.87	2.84	2.81	2.78	2.75	2.72	2.71	2.68	2.62	2.59	2.58	2.57	18
19	8.19	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.36	3.30	3.24	3.19	3.15	3.12	3.08	3.05	3.03	3.00	2.96	2.92	2.89	2.87	2.84	2.80	2.76	2.73	2.71	2.67	2.65	2.63	2.60	2.55	2.51	2.50	2.49	19
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.29	3.23	3.18	3.13	3.09	3.05	3.02	2.99	2.96	2.94	2.90	2.86	2.83	2.80	2.78	2.73	2.69	2.67	2.64	2.61	2.58	2.56	2.54	2.48	2.44	2.43	2.42	20
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.18	3.12	3.07	3.02	2.98	2.94	2.91	2.88	2.85	2.83	2.78	2.75	2.72	2.69	2.67	2.62	2.58	2.55	2.53	2.50	2.47	2.45	2.42	2.36	2.33	2.32	2.31	22
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.09	3.03	2.98	2.93	2.89	2.85	2.82	2.79	2.76	2.74	2.70	2.66	2.63	2.60	2.58	2.53	2.49	2.46	2.44	2.40	2.38	2.36	2.33	2.27	2.24	2.22	2.21	24
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09	3.02	2.96	2.90	2.86	2.82	2.78	2.75	2.72	2.69	2.66	2.62	2.58	2.55	2.53	2.50	2.45	2.42	2.39	2.36	2.33	2.30	2.28	2.25	2.19	2.16	2.14	2.13	26
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03	2.96	2.90	2.84	2.79	2.75	2.72	2.68	2.65	2.63	2.60	2.56	2.52	2.49	2.46	2.44	2.39	2.35	2.32	2.30	2.26	2.24	2.22	2.19	2.13	2.09	2.08	2.07	28
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.91	2.84	2.79	2.74	2.70	2.66	2.63	2.60	2.57	2.55	2.51	2.47	2.44	2.41	2.39	2.34	2.30	2.27	2.25	2.21	2.18	2.16	2.13	2.07	2.03	2.02	2.01	30
35	7.42	5.27	4.40	3.91	3.59	3.37	3.20	3.07	2.96	2.88	2.80	2.74	2.69	2.64	2.60	2.56	2.53	2.50	2.47	2.44	2.40	2.36	2.33	2.31	2.28	2.23	2.19	2.16	2.14	2.10	2.07	2.05	2.02	1.96	1.92	1.90	1.89	35
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.73	2.66	2.61	2.56	2.52	2.48	2.45	2.42	2.39	2.37	2.33	2.29	2.26	2.23	2.20	2.15	2.11	2.08	2.06	2.02	1.99	1.97	1.94	1.87	1.83	1.82	1.81	40
45	7.23	5.11	4.25	3.77	3.45	3.23	3.07	2.94	2.83	2.74	2.67	2.61	2.55	2.51	2.46	2.43	2.39	2.36	2.34	2.31	2.27	2.23	2.20	2.17	2.14	2.09	2.05	2.02	2.00	1.96	1.93	1.91	1.88	1.81	1.77	1.75	1.74	45
50	7.17	5.06	4.20	3.72	3.41	3.19	3.02	2.89	2.79	2.70	2.63	2.56	2.51	2.46	2.42	2.38	2.35	2.32	2.29	2.27	2.22	2.18	2.15	2.12	2.10	2.05	2.01	1.97	1.95	1.91	1.88	1.86	1.82	1.76	1.71	1.70	1.69	50
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.56	2.50	2.44	2.39	2.35	2.31	2.28	2.25	2.22	2.20	2.15	2.12	2.08	2.05	2.03	1.98	1.94	1.90	1.88	1.84	1.81	1.78	1.75	1.68	1.63	1.62	1.60	60
70	7.01	4.92	4.07	3.60	3.29	3.07	2.91	2.78	2.67	2.59	2.51	2.45	2.40	2.35	2.31	2.27	2.23	2.20	2.18	2.15	2.11	2.07	2.03	2.01	1.98	1.93	1.89	1.85	1.83	1.78	1.75	1.73	1.70	1.62	1.57	1.56	1.54	70
80	6.96	4.88	4.04	3.56	3.26	3.04	2.87	2.74	2.64	2.55	2.48	2.42	2.36	2.31	2.27	2.23	2.20	2.17	2.14	2.12	2.07	2.03	2.00	1.97	1.94	1.89	1.85	1.82	1.79	1.75	1.71	1.69	1.65	1.58	1.53	1.51	1.50	80
100	6.90	4.82	3.98	3.51	3.21	2.99	2.82	2.69	2.59	2.50	2.43	2.37	2.31	2.27	2.22	2.19	2.15	2.12	2.09	2.07	2.02	1.98	1.95	1.92	1.89	1.84	1.80	1.76	1.74	1.69	1.66	1.63	1.60	1.52	1.47	1.45	1.43	100
200	6.76	4.71	3.88	3.41	3.11	2.89	2.73	2.60	2.50	2.41	2.34	2.27	2.22	2.17	2.13	2.09	2.06	2.03	2.00	1.97	1.93	1.89	1.85	1.82	1.79	1.74	1.69	1.66	1.63	1.58	1.55	1.52	1.48	1.39	1.33	1.30	1.28	200
500	6.69	4.65	3.82	3.36	3.05	2.84	2.68	2.55	2.44	2.36	2.28	2.22	2.17	2.12	2.07	2.04	2.00	1.97	1.94	1.92	1.87	1.83	1.79	1.76	1.74	1.68	1.63	1.60	1.57	1.52	1.48	1.45	1.41	1.31	1.23	1.20	1.17	500
1000	6.66	4.63	3.80	3.34	3.04	2.82	2.66	2.53	2.43	2.34	2.27	2.20	2.15	2.10	2.06	2.02	1.98	1.95	1.92	1.90	1.85	1.81	1.77	1.74	1.72	1.66	1.61	1.58	1.54	1.50	1.46							